

LEIGH ANNE MILLER

FORWARD DEPLOYED ENGINEER · AI SOLUTION ARCHITECT

PALO ALTO, CA · DENVER, CO

leighmillerengineering@gmail.com

(650) 469-6032

linkedin.com/in/leighannemillerengineering

github.com/leighanne77

I'm a forward-deployed engineer: I embed with the customer, find the real problem, and **ship the AI system to production myself.**

\$40B/YR

AMPLIFY, HP's first API-based channel data exchange.
Led enterprise data-strategy programs targeting \$15B in new revenue.

0 → 1

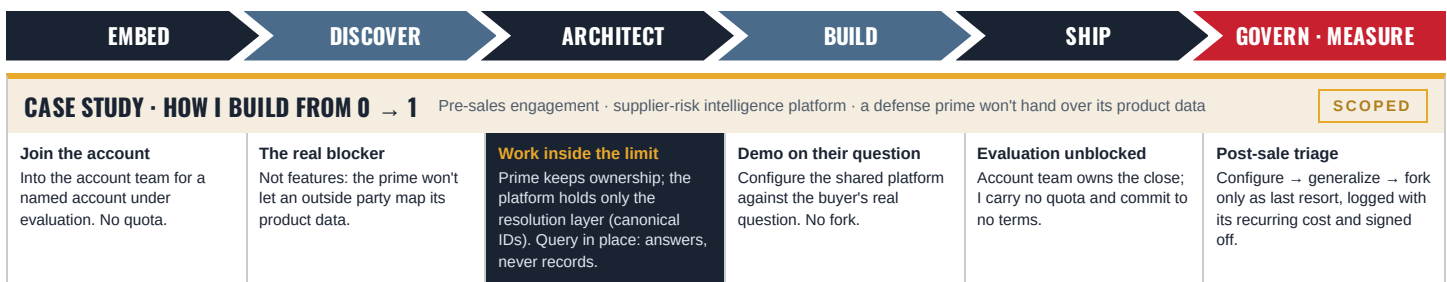
Multiple products opened from zero: discovery, demos, Tier-1 deployment, and the engineering.

\$300M

Unified data platform restarted and shipped at Nielsen/Gracenote after a five-year impasse. I supplied the revenue case.

THE FDE LOOP

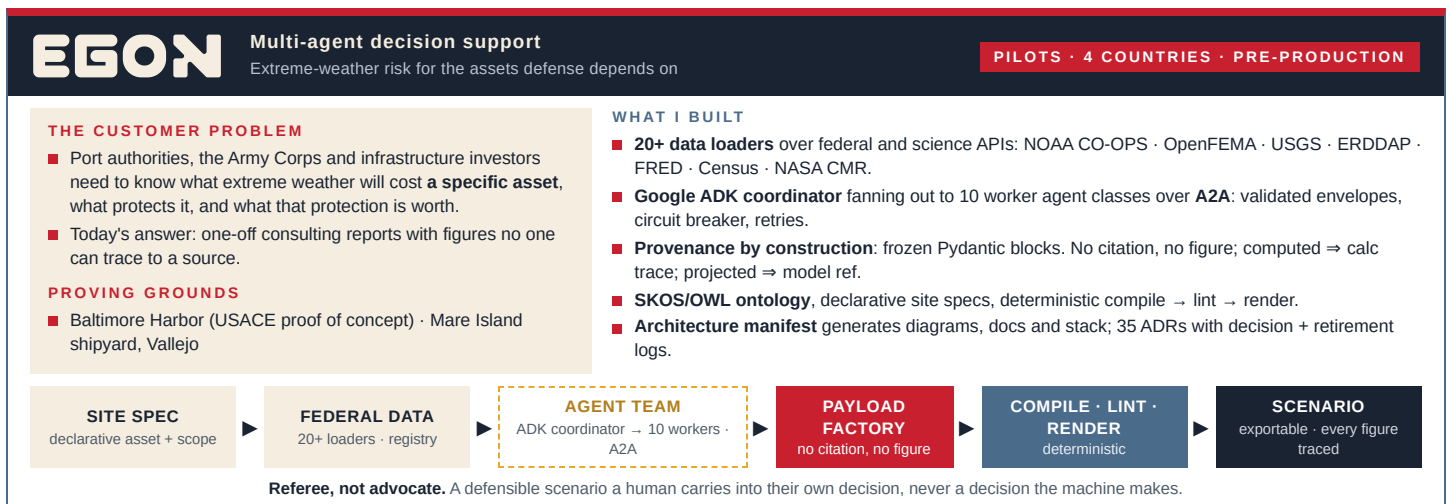
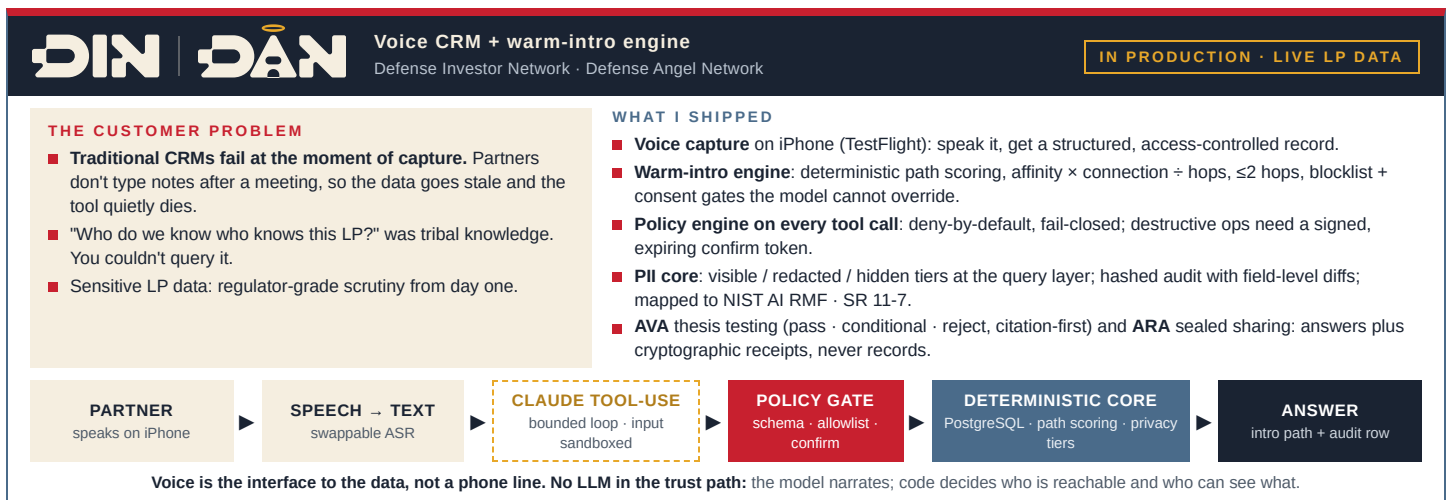
HOW EVERY ENGAGEMENT RUNS · FIELD FEEDBACK + EVALS FEED THE NEXT LOOP



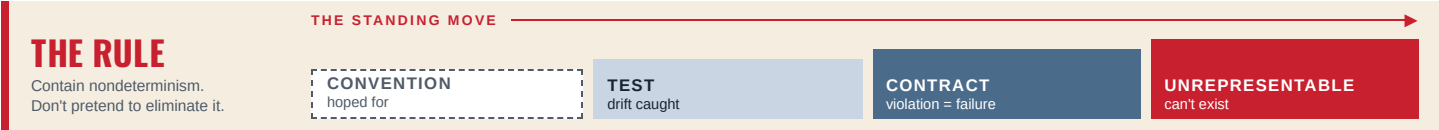
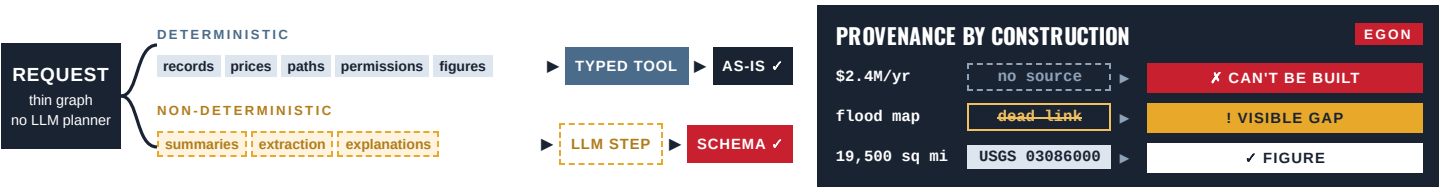
FDE WORK: LIVE, IN PRODUCTION

BUILT SOLO · DIN · DAN IN PRODUCTION · EGON IN PILOTS

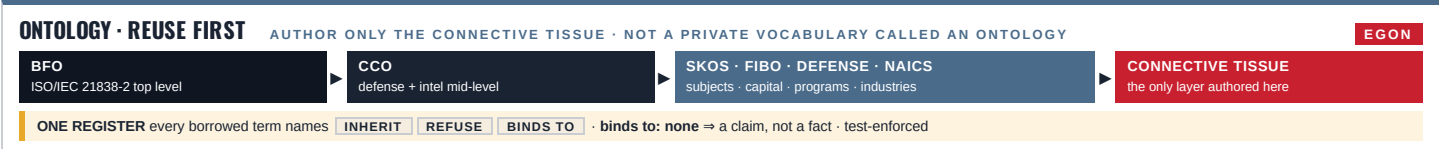
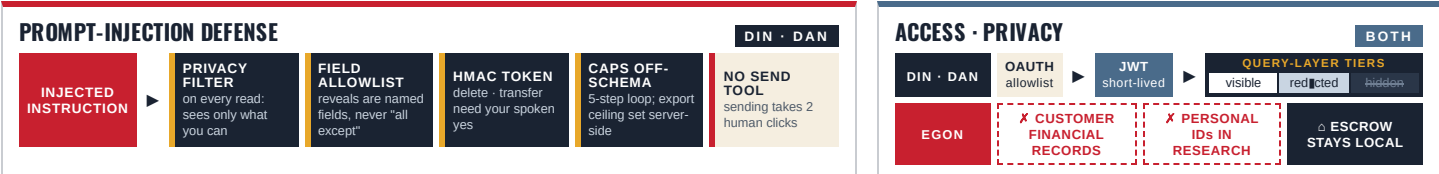
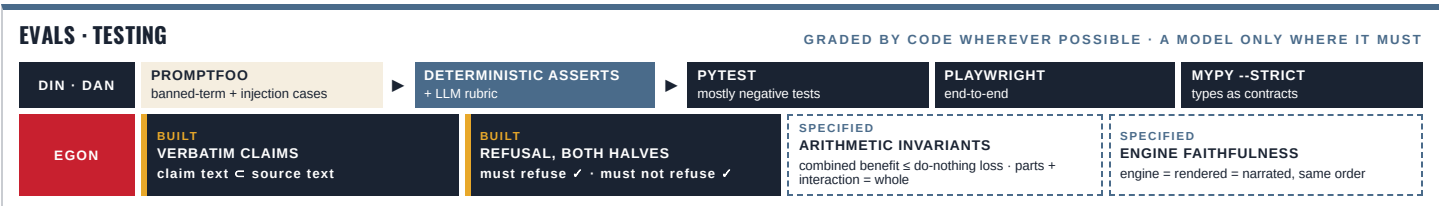
I OWNED, SOLO **Discovery** **Architecture** **Full stack** **ML ops** **Multi-tenant cloud** **Security** **Governance** **Rollout**



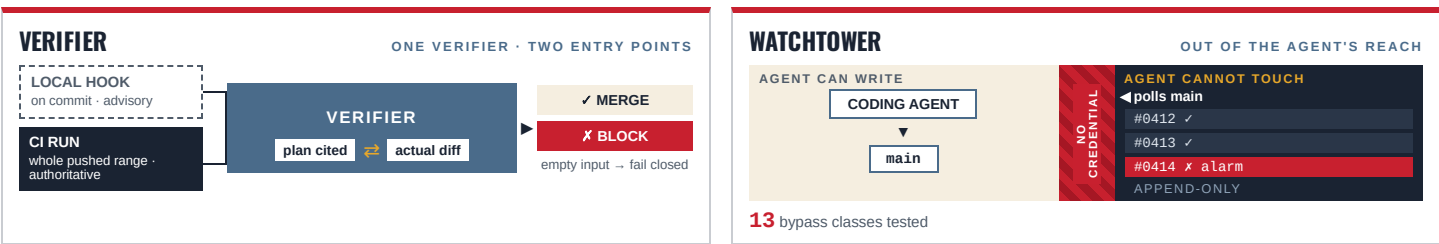
HOW I BUILD ENFORCED CONTRACTS · LIGHT DECLARATIVE ORCHESTRATION



PRODUCTION PATTERNS AND WHERE EACH ONE RUNS



WATCHTOWER BESPOKE TOOLING TO GOVERN AI AGENTS ON REGULATED, DATA-INTENSIVE BUILDS



FIELD RECORD CUSTOMER-FACING DEPLOYMENTS

2018-21 HP AMPLIFY: partner-facing channel data exchange, API-first \$40B/YR	2021-23 Nielsen / Gracenote Unified data platform on AWS, uninstalled after a five- year impasse \$300M/YR	2023-24 Ripley Decision Advantage Founder-engineer: compound AI systems, built and shipped solo	2024 SpinF AI · BootstrapLabs Real-time voice AI on live sales calls, discovery through Tier-1 deployment 0 → 1	2025-26 X, The Moonshot Factory AI engineer	2026- DIN · DAN AI Solution Architect. Voice CRM + intro engine on live LP data LIVE	2026- Ripley · EGON Multi-agent decision support for ports and the Army Corps PILOTS
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THE STACK WHAT I BUILD WITH

LANGUAGES	Python · TypeScript · Java · SQL
LLMS · AGENTS	Claude · Llama (local) · Claude Code · Google ADK · A2A · MCP · LangGraph
CONTRACTS · TESTS	Pydantic v2 (frozen models) · mypy --strict · pytest · Playwright · promptfoo
DATA	PostgreSQL/pgvector · SQLAlchemy · BigQuery · rdfliib (SKOS/OWL) · Kafka
CLOUD · APP	GCP (Cloud Run · Cloud SQL · Vertex AI) · Confidential Space · AWS · FastAPI · React · Capacitor iOS
VOICE · TRACING	Google Chirp · ElevenLabs · Deepgram · Groq · Arize · OpenTelemetry

Q WHAT'S UNDER THE VOICE INTERFACE?

DIN · DAN



Q AN EXTERNAL SERVICE IS SLOW OR FAILING?

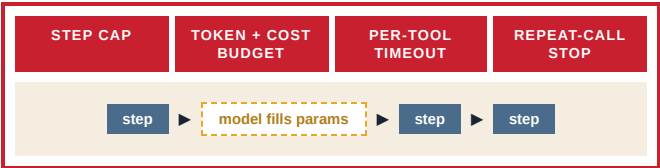
Never a 30-second spinner. Never stale data shown as fresh.



No retry on a bad 400 · idempotency keys on writes · long jobs go async: job ID, then poll or push.

Q HOW DO YOU STOP RUNAWAY TOOL LOOPS?

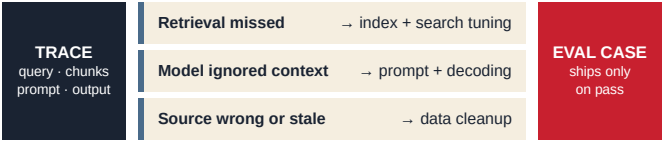
The graph decides what runs next. The model only fills in parameters.



A limit hit stops the run and escalates to a human, with the full trace.

Q THE SYSTEM GAVE A WRONG ANSWER. NOW WHAT?

Classify the failure first. Each cause has a different fix.



Q AUTHENTICATION AND AUTHORIZATION?

Identity is short-lived; access is checked on every request and logged.



Services: workload identity, mTLS, rotated secrets. Google OAuth allowlist → JWT; privacy tiers at the query layer.

Q HOW HANDS-ON ARE YOU?

I code daily, Python first. Don't take my word for it:

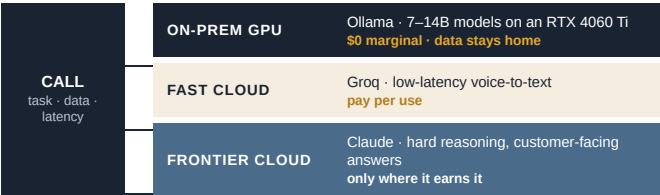
github.com/leighanne77/DESS
Public tree of the DIN · DAN CRM + intro engine. Runs end-to-end on fictional data.

github.com/leighanne77/MAS_Example
Provenance by construction: the EGON trust-core pattern.

\$ git clone ... && pytest · FastAPI · LangGraph · ADK · Node · Java

Q HOW DO YOU MANAGE MODEL COSTS?

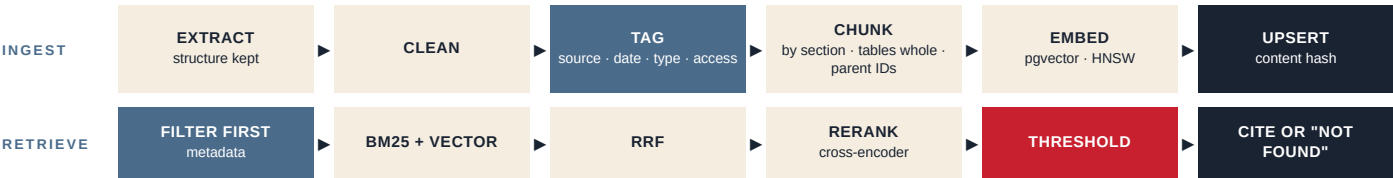
Route each call to the cheapest place that meets the bar, and meter everything.



On-prem runs coding assist, the local Llama judge and bulk jobs. per-user token + voice budgets; web-search spend capped and audited. Right-size infra too: PostgreSQL, not Spark.

Q RAG: INGESTION, CHUNKING, FEWER HALLUCINATIONS?

Permissions attach at ingest. Security can't be bolted on later.



Measured, not eyeballed: a labeled eval set, recall@k and faithfulness. Every request logged to tell a retrieval miss from a generation miss.